Jens Einar Bremnes

PhD · Marine Cybernetics, NTNU

Research Scientist & Engineer · GeoNord AS

Applied researcher and engineer in marine robotics and autonomous systems, with experience across academia, defence, and industry. My work spans autonomy, risk modeling, planning, control and navigation, with experimental validation in real-world environments.

✉ Email Google Scholar GitHub ORCID LinkedIn ResearchGate

132	7	8	8
CITATIONS	H-INDEX	JOURNAL PAPERS	CONFERENCE PAPERS

RESEARCH INTERESTS

- **Autonomy for robotics** — planning, decision-making, adaptive behaviors, mission execution and supervision
- **Risk management in robotics** — risk analysis, probabilistic AI (Bayesian methods, Gaussian processes), supervisory control, and formal safety guarantees
- **Field robotics** — Real-world implementation and deployments, including Arctic operations
- **Multi-vehicle coordination** — collaborative control and communication for multiple autonomous systems
- **Ocean mapping and monitoring** — sampling strategies for acoustic, optical and chemical sensing
- **Sensor and data fusion** — integration of data from multiple sensors and data sources for navigation and situational awareness

EDUCATION

PhD, Marine Technology (Marine Cybernetics)

2019 - 2024

Norwegian University of Science and Technology (NTNU), Trondheim, Norway

Thesis: *Safe Autonomy in Marine Robotics*

PhD candidate in the UNLOCK project at the Dept. of Marine Technology. Affiliated with the Centre for Autonomous Marine Operations and Systems (NTNU AMOS) and the Nansen Legacy project. Research focused on risk modelling, risk control and autonomous planning and decision-making for marine robotics operating in unstructured environments.

Exchange studies: Spring 2020 — Dept. of EECS, UC Berkeley, USA (reachability analysis, Gaussian processes, model predictive control); Fall 2019 — UNIS, Longyearbyen, Svalbard (doctoral course *AT-834 Arctic Marine Measurement Techniques, Operations and Transport*).

MSc, Marine Technology (Marine Cybernetics)

2014 - 2019

Norwegian University of Science and Technology (NTNU), Trondheim, Norway

Thesis: *Towards Robust Autonomy of Underwater Vehicles in Arctic Operations*

GPA: 4.3 (B+). Coursework in advanced control systems, robotics, risk analysis, artificial intelligence, hydrodynamics, and programming. Began an integrated PhD programme in the fifth year.

Exchange studies: Spring 2018 — Dept. of ECE, *Instituto Superior Técnico (IST)*, Lisbon (robotics, optimisation, control theory, offshore wind energy).

EXPERIENCE

Sept 2025 – present

Engineer

GeoNord AS · Alta, Norway

Seabed and terrain mapping: field data collection with multibeam echo sounders and laser scanners from vessel and drone; processing and analysis of point clouds, 3D models, orthophotos, contour maps, and volume calculations. Survey planning, data acquisition, and post-processing for marine, coastal and construction projects.

May 2023 – Sept 2025

Scientist

Norwegian Defence Research Establishment (FFI) · Kjeller, Norway

Research and development on autonomy for HUGIN-class AUVs. Developed novel algorithms for autonomous mission planning and decision-making; software development in C++ (primary), Python, and MATLAB with integration on operational AUV systems and operator interfaces. Field campaigns aboard research vessel *H.U. Sverdrup II*. Scientific dissemination via journal articles, conference papers, and reports; contributed to research projects, proposals and master's student supervision.

Aug 2019 – May 2023

PhD Fellow

NTNU Department of Marine Technology · Trondheim, Norway

Autonomy research in marine robotics with focus on risk management, AI, and control systems. Contributed to the Applied Underwater Robotics Laboratory (AUR-Lab): operation and maintenance of AUV, USV, and ROV fleets; scientific cruises in the North Sea, Svalbard, Lake Mjøsa, and Trøndelag fjords. Visiting Scholar at UC Berkeley (EECS, Spring 2020). Co-developed TMR06 Autonomous Marine Systems; scientific assistant for TMR4240 Marine Control Systems; co-supervised master's students.

Aug 2014 – May 2019

Student Assistant & Recruitment Assistant

NTNU Department of Marine Technology · Trondheim, Norway

Student assistant in TMR4240 Marine Control Systems (Autumn 2018) and TMR4160 Computer Methods for Marine Applications (Spring 2019). Recruited high school students to the Master's programme in Marine Technology.

June – Aug 2018

Summer Intern

DNV GL, Digital Assurance · Trondheim, Norway

Software development in C++ and Python with ROS for the *ReVolt* autonomous ship testbed; vessel maintenance.

June – Aug 2017

Summer Intern

Equinor, New Energy Solutions · Stavanger, Norway

Monopile design and parametric studies for large-scale (10+ MW) offshore wind turbines; structural analysis in ULS and FLS.

RESEARCH PROJECTS

2023 – 2025

CsasPlanner

Norwegian Defence Research Establishment (FFI) · Lead developer

FFI project on decisional autonomy for circular synthetic aperture sonar (CSAS) surveying. Developed autonomous mission planning strategies for AUV-based CSAS operations. Software implementation and experimental validation on the HUGIN AUV.

2023 – 2025

Intelligent ID

Norwegian Defence Research Establishment (FFI) · Lead developer

FFI project on autonomous optical object identification using AUVs. Developed replanning algorithms for robust optical inspection using in-situ sonar data. Software implementation and experimental validation on the HUGIN AUV.

2023 – 2025

SmartAUVs

Research Council of Norway / Plymouth Marine Laboratory · Collaborator (FFI)

Smart AUVs for Detection and Quantification of Greenhouse Gas Seepage in the Oceans. Developed autonomy algorithms for adaptive sampling of CO_2 and methane seepage triggered by real-time detections. Software implementation and experimental validation on the HUGIN AUV.

2021 – 2024

The Nansen Legacy

Research Council of Norway / UiT – The Arctic University of Norway · Participant

Affiliated with the Nansen Legacy project, a large research project on the Arctic marine ecosystems. Participated in the Q2 2021 seasonal cruise in the Arctic Ocean, contributing to ROV-based data collection, under-ice operations, and bio-optical measurements.

2018 – 2024

UNLOCK

Research Council of Norway, FRINATEK / NTNU · PhD Candidate

PhD was funded by the UNLOCK project (Unlocking the Potential of Autonomous Systems and Operations through Supervisory Risk Control). The project involved online risk modelling, testing, and verification of safe autonomous systems, with case studies on AUVs for ocean mapping and UAVs for industrial inspection. My work focused on supervisory risk control and safe autonomy for marine robotics. Hosted at the Dept. of Marine Technology, NTNU; affiliated with SFF AMOS.

PUBLICATIONS

JOURNAL ARTICLES

1. Summers, **Bremnes** et al. Assessing Arctic under sea-ice light regimes and ice algal bio-optical properties using ROV-based hyperspectral imaging. *Polar Biology*, 2026.
2. Kristensen, Maidana, Utne, **Bremnes** Evaluating the effect of risk metrics for supporting operational decision-making by autonomous surface vehicles. *Ocean Engineering*, 2025. 2 cited
3. **Bremnes**, Utne, Krogstad, Sørensen Holistic risk modeling and path planning for marine robotics. *IEEE Journal of Oceanic Engineering*, 2024. 6 cited
4. **Bremnes**, Fyrvik, Krogstad, Sørensen Design of a Switching Controller for Tracking AUVs with an ASV. *IEEE Transactions on Control Systems Technology*, 2024. 10 cited
5. Vasilijevic, **Bremnes**, Ludvigsen Remote operation of marine robotic systems and next-generation multi-purpose control rooms. *Journal of Marine Science and Engineering*, 2023. 16 cited
6. Yang, **Bremnes**, Utne Online risk modeling of autonomous marine systems: Case study of autonomous operations under sea ice. *Ocean Engineering*, 2023. 21 cited
7. **Bremnes**, Brodtkorb, Sørensen Hybrid observer concept for sensor fusion of sporadic measurements for underwater navigation. *International Journal of Control, Automation and Systems*, 2020. 5 cited
8. **Bremnes**, Thieme, Sørensen, Utne, Norgren A Bayesian approach to supervisory risk control of AUVs applied to under-ice operations. *Marine Technology Society Journal*, 2020. 24 cited

CONFERENCE PAPERS

9. **Bremnes**, Krogstad Planning in circles: Autonomous planning for circular synthetic aperture sonar surveying with AUVs. *UDT*, 2025. ★ Best Paper Award
10. **Bremnes**, Rørstad, Krogstad, Midtgaard Adaptive AUV Planning For Robust Optical Object Inspection Using In-Situ Sonar Data. *AUV Symposium*, 2024. 1 cited
11. Vasilijevic, Gravråkmo, Barstein, **Bremnes** Infrastructure for remote experimentation in the Trondheim fjord. *OCEANS*, 2023. 7 cited
12. Yang, **Bremnes**, Utne A system-theoretic approach to hazard identification of operation with multiple autonomous marine systems (AMS). *ESREL*, 2022. 6 cited
13. Fyrvik, **Bremnes**, Sørensen Hybrid tracking controller for an ASV providing mission support for an AUV. *CAMS*, 2022. 7 cited
14. **Bremnes** et al. Optimization-based planning and control of AUVs applied to adaptive sampling under ice. *AUV Symposium*, 2020. 2 cited
15. **Bremnes** et al. Intelligent risk-based under-ice altitude control for autonomous underwater vehicles. *OCEANS*, 2019. 20 cited
16. **Bremnes**, Brodtkorb, Sørensen Sensor-based hybrid translational observer for underwater navigation. *CAMS*, 2019. 3 cited

REPORTS

17. **Bremnes**, Krogstad CsasPlanner: Decisional autonomy for circular synthetic aperture sonar surveying with AUVs. *FFI report, exempt from public access*, 2025.
18. **Bremnes** et al. Intelligent ID: Replanning for robust optical identification with HUGIN using EM2040. *FFI report, exempt from public access*, 2025.
19. Ludvigsen, Assmy, Adams, **Bremnes** et al. Seasonal cruise Q2 2021: Cruise report. *The Nansen Legacy Report Series*, 2022. 2 cited
20. Metlay, St. Clair, Twomey, **Bremnes**, Rødseth, Massaiu Autonomous transportation technologies: Society and individuals in the loop. Report of the discussions of breakout session. *Proceedings of the First International Workshop on Autonomous Systems Safety*, 2019.

THESES

21. Bremnes Safe Autonomy in Marine Robotics. *PhD thesis, Dept. of Marine Technology, NTNU*, 2024.
22. Bremnes Towards Robust Autonomy of Underwater Vehicles in Arctic Operations. *MSc thesis, Dept. of Marine Technology, NTNU*, 2019.

TEACHING

CODE	COURSE	ROLE	YEAR(S)
TMR06	Autonomous Marine Systems	Course co-developer; contributed to syllabus design, lecture notes, and project specification; lecturing, and grading	2020
TMR4240	Marine Control Systems	Student assistant (2018), scientific assistant (2019-2020); exercise supervision, grading, and student guidance	2018–2020
TMR4160	Computer Methods for Marine Applications	Student assistant	2019

MASTER'S STUDENT SUPERVISION

- **Thomas Erling Lone** (co-supervisor) · MSc, NTNU Department of Marine Technology, 2020
  Navigation Techniques for Underwater Vehicles in Polar Regions
- **Torbjørn Reitan Fyrvik** (co-supervisor) · MSc, NTNU Department of Marine Technology, 2022
  Methods for Ocean Mapping with Combined ASV and AUV Platforms
- **Hans Petter Vatne** (co-supervisor) · MSc, NTNU Department of Engineering Cybernetics, 2025
  Motion Planning for the Hugin 1000 AUV

OTHER

- **Substitute teacher**, mathematics and natural science, grades 8–10 · Sandfallet Secondary School, Alta (Jan 2016, 2017, 2018)
- **Science mentor**, ENT3R, Trondheim (2015–2017) — weekly tutoring of homework and hands-on science experiments for secondary school students

AWARDS & MEDIA OUTREACH

2025 **Best Paper Award**, Underwater Defence Technology (UDT) 2025, Oslo
Planning in circles: Autonomous planning for circular synthetic aperture sonar surveying with AUVs (certificate)

MEDIA

forskning.no · Nov 2024	Han vil lære undervannsroboter å frykte døden
gemini.no · Jun 2022	Forskere undersøkte alt, overalt, på samme tid
sciencenorway.no · May 2021	Mixing production deep into the ocean

SOFTWARE PROJECTS

Python github.com/jensbremnes/geobn ↗

geobn

Python library for Bayesian network inference over geospatial data. Combines heterogeneous data sources (GeoTIFF, WCS, in-memory arrays) to generate insight over geographical areas.

TECHNICAL SKILLS

Programming	C++, Python, MATLAB / Simulink
Frameworks & tools	ROS (Robot Operating System), Git, Docker, Linux, GIS
Methods	Control theory, sensor fusion, Gaussian processes, Bayesian inference, risk analysis, autonomous planning, data visualisation